

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

10 Week: 2 May 2017 / 2 Mei 2017

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P9.48 $\vec{r}_{\text{CM}} = (11.7\hat{i} + 13.3\hat{j})$ cm

P9.50 $\vec{r}_{\text{CM}} = (-0.0067\hat{j})$ nm if the oxygen atom is at the origin.

P9.52 (a) Sketch (b) $\vec{r}_{\text{CM}} = (-2.00\hat{i} - 1.00\hat{j})$ m (c) $\vec{v}_{\text{CM}} = (3.00\hat{i} - 1.00\hat{j})$ m/s

(d) $\vec{P} = (15.0\hat{i} - 5.00\hat{j})$ kg·m/s

P10.4 144 rad

P10.10 The disk’s angular acceleration is not constant as stated.

P10.18 (a) 0.605 m/s (b) 17.3 rad/s (c) 5.82 m/s (d) The length of the pedal cranks.

P10.27 -3.55 N·m

P10.69 (a) 12.5 rad/s (b) $\Delta\theta = 128$ rad

P10.72(a) (i) -794 N·m (ii) -2510 N·m (iii) 0 N·m (iv) -1160 N·m (v) 2940 N·m

You only need to do P10.72(a)

P11.3 (a) $7.00\hat{k}$ (b) 60.3°

P11.6 (a) 168° (b) 168° or 11.9° (c) Method (a).

P11.9 (a) $|\vec{F}_3| = |\vec{F}_1| + |\vec{F}_2|$ (b) Explanation